Batch Effect Management

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Objectives of this workshop

- To recognise the importance of addressing batch effects for research reproducibility.
- To understand assumptions, applications and limitations of existing methods handling batch effects.
- To gain practical skills in managing batch effects and evaluating correction effectiveness.

Content

- ➢ Background on batch effects
- ➢ Case studies
- ➢ Workflow for batch effect management
  - Batch effect detection
  - Managing batch effects
  - Assessing batch effect correction
- ➢ Conclusions and take-home messages

Batch effects

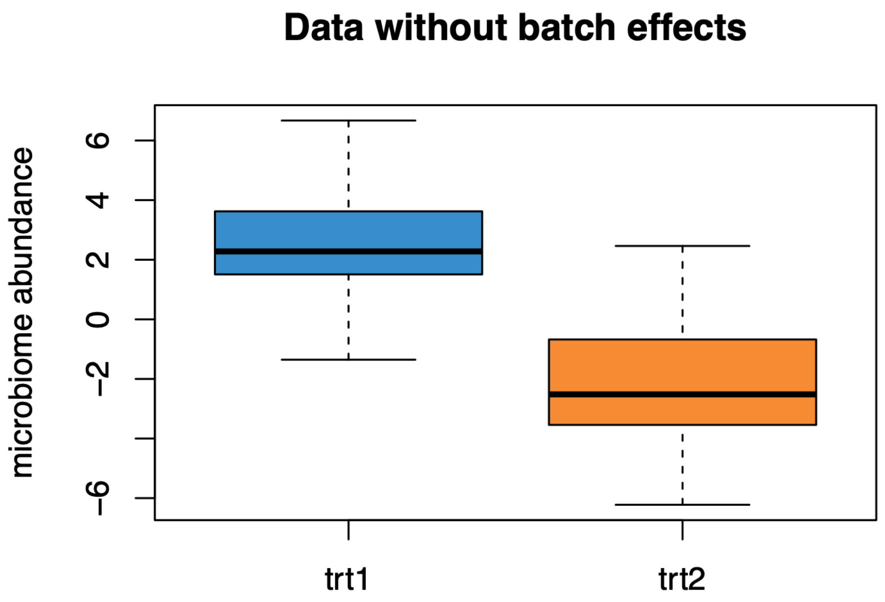
Definition: A batch effect is unwanted variation in measured data arising from biological, technical or computational factors that are not the primary focus of the study and that may obscure the biological effects of interest (e.g., treatment effects).

- If a batch factor is associated with the outcome of interest, it may act as a confounder.
- If it is independent of the outcome, it is not a confounder, but it may still obscure the signal and reduce statistical power.
- This workshop focuses only on categorical batch factors.

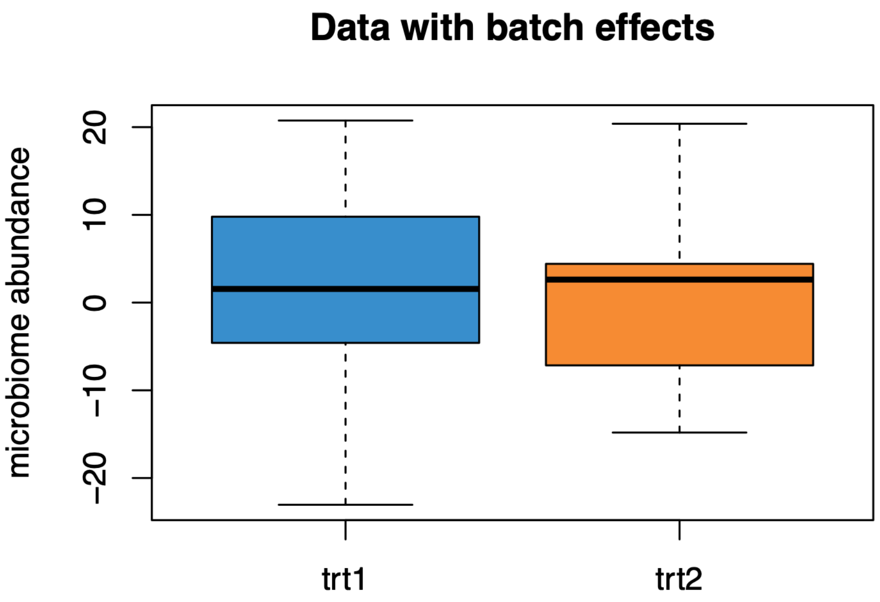
Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

Batch effects

Consequence:



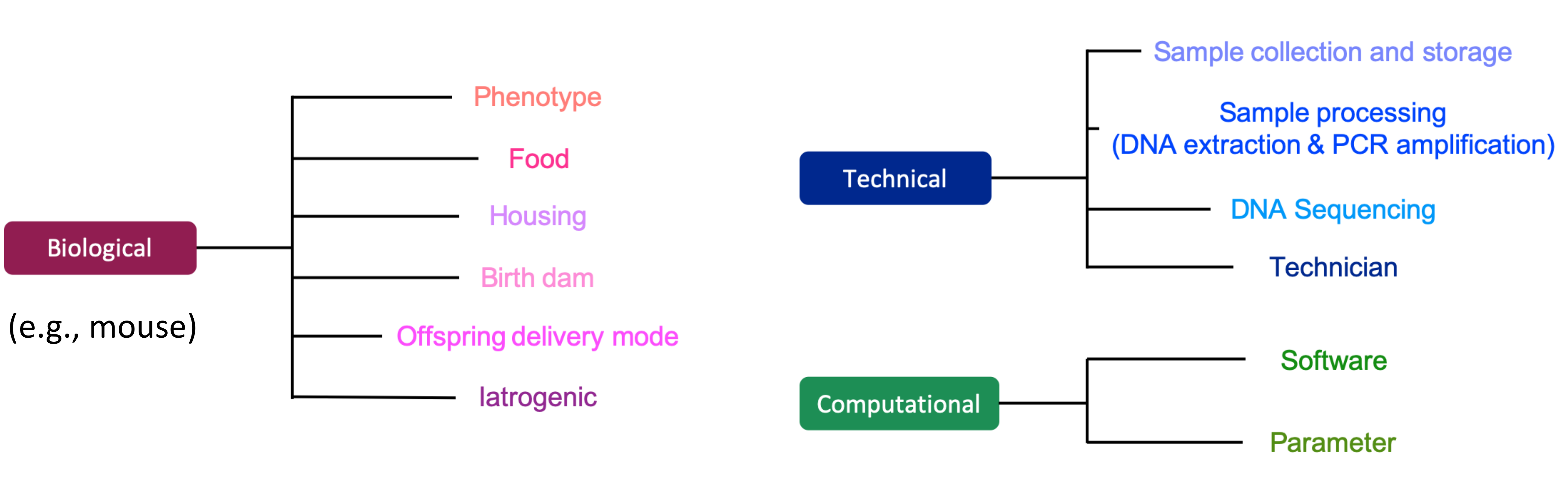
P < 0.001 of the treatment effect in T-test



P > 0.05 of the treatment effect in T-test

Potential batch sources

Batch effects may happen in any step of experiments.



Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

Assumptions about batch effects

Many statistical methods that correct for batch effects assume balanced batch x treatment designs, which means the batch and treatment factors are independent.

Balanced

	Treat 1	Treat 2
Batch 1	10	10
Batch 2	10	10

Unbalanced (confounding)

	Treat 1	Treat 2
Batch 1	4	16
Batch 2	16	4

Nested (confounding)

	Treat 1	Treat 2
Batch 1	10	0
Batch 2	0	10
Batch 3	10	0
Batch 4	0	10

Nested ✕

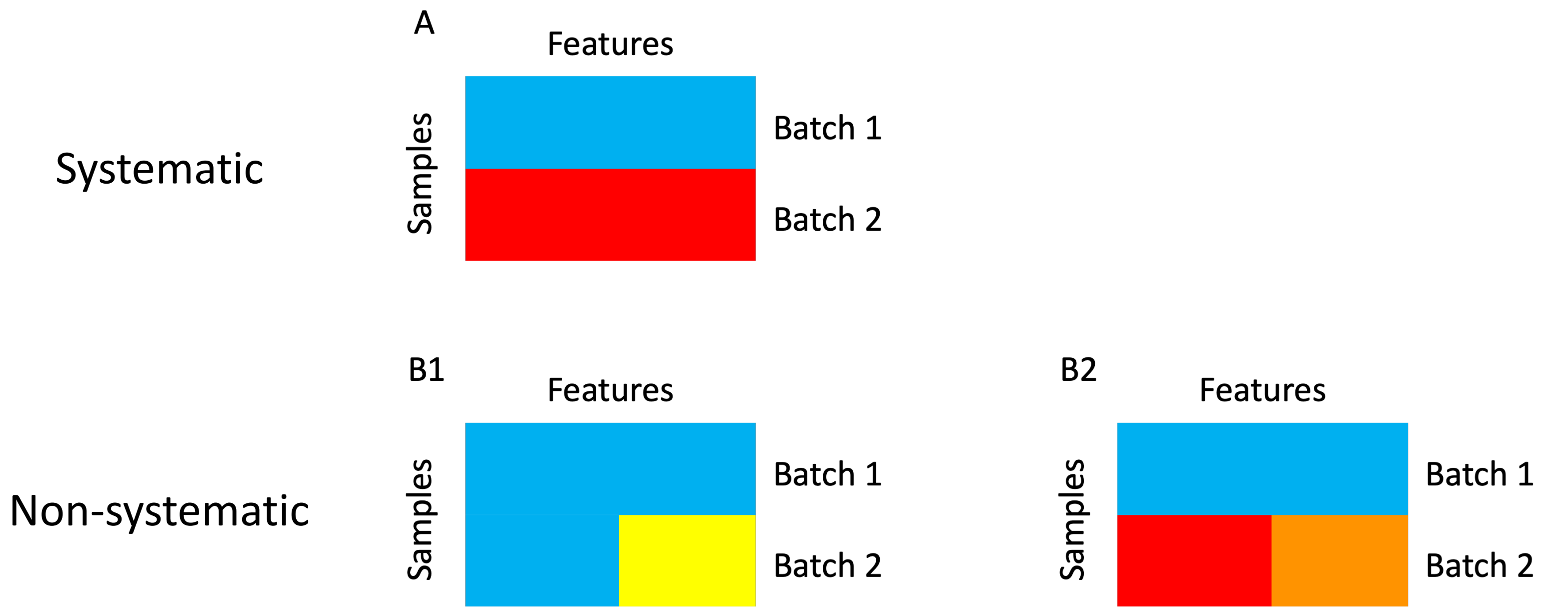
	Treat 1	Treat 2
Batch 1	0	20
Batch 2	20	0

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

Assumptions about batch effects

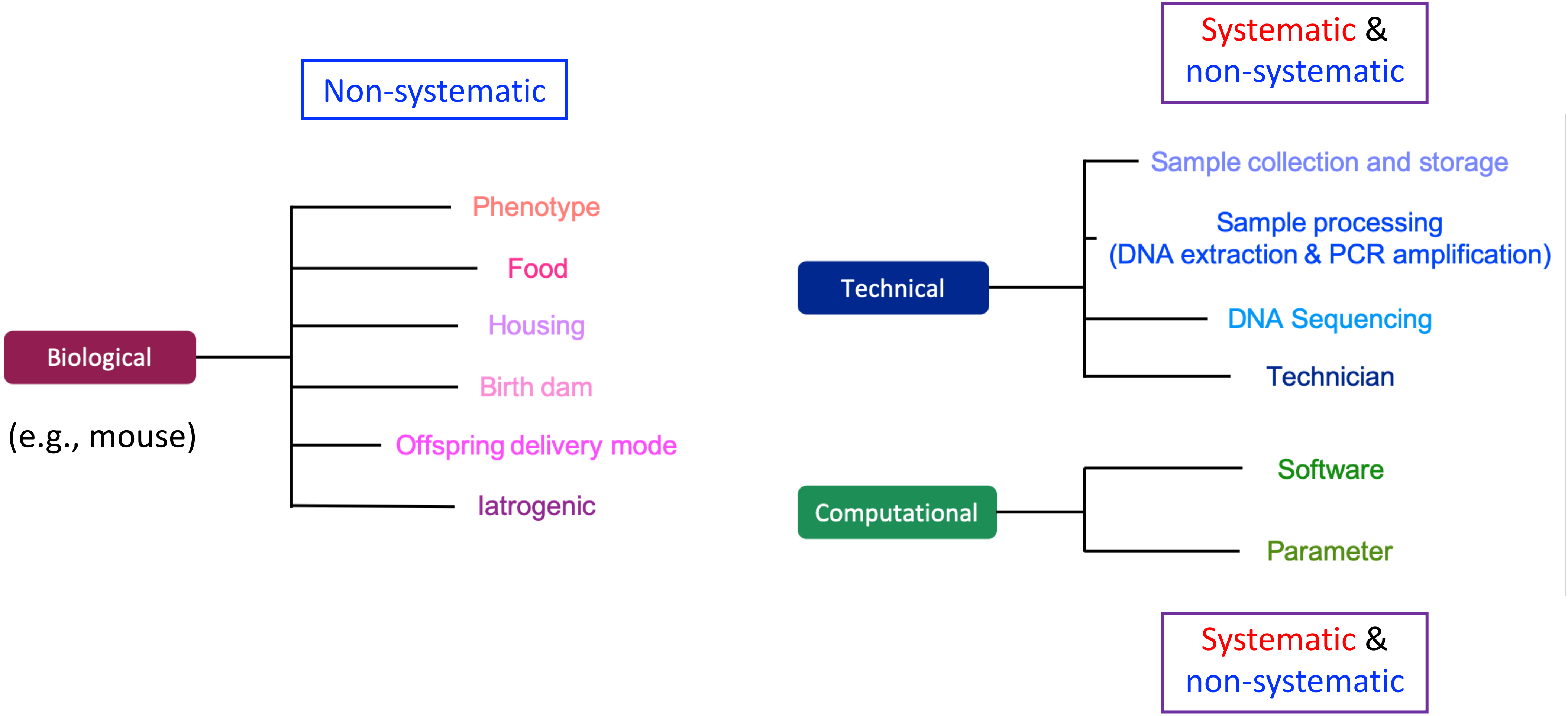
Batch effects can influence variables differently:

- Systematic batch effects exhibit a shared pattern across all variables, although the variable-specific effects are not necessarily identical.
- Non-systematic batch effects have heterogeneous influences across variables. They may affect only a subset of variables or affect different groups of variables with different magnitudes or directions.



Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

Assumptions about batch effects

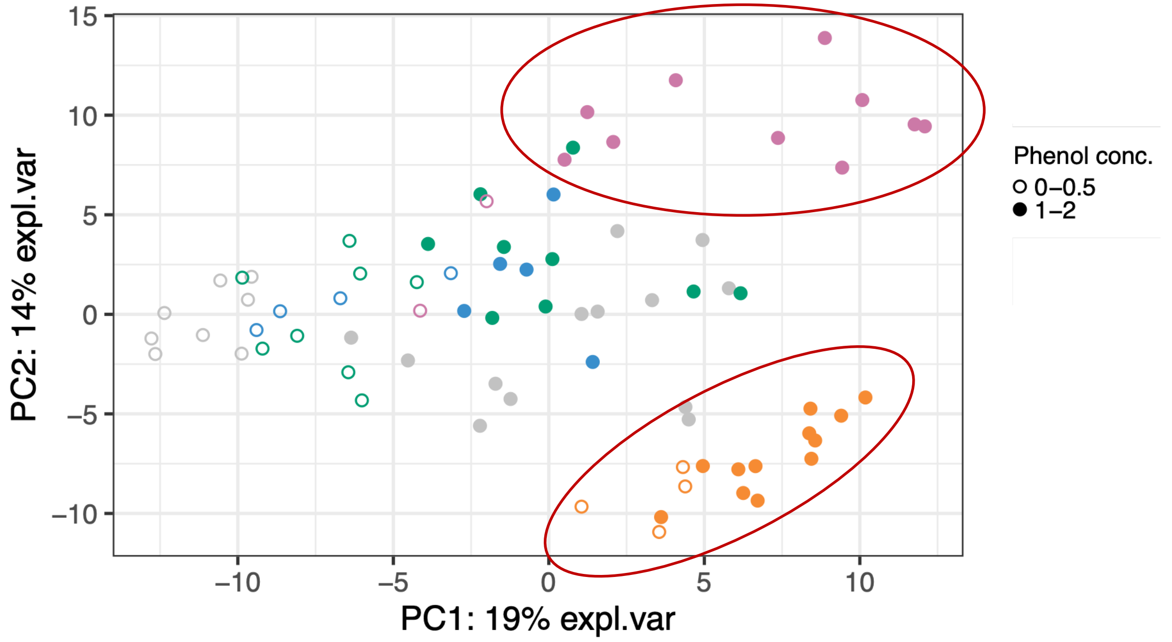


Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

Case studies

Anaerobic Digestion (AD data): ★

- Bioreactor experiment: aimed at improving biowaste digestion
- 567 microbial variables & 75 samples
- Treatment effect: 2 levels of phenol concentrations
- (Technical) batch effect: samples processed on 5 different dates



Chapleur, et al (2016). Increasing concentrations of phenol progressively affect anaerobic digestion of cellulose and associated microbial communities. *Biodegradation*.

Case studies

Sponge data:

- Investigating differences in microbial composition between specific sponge tissues
- 24 microbial variables & 32 samples
- Effect of interest: 2 different tissues
- (Technical) batch effect: sample processed on 2 different experimental gels
- Data characteristic: completely balanced design

	Tissue 1	Tissue 2
Batch 1	8	8
Batch 2	8	8

Sacristan-Soriano, et al (2011). Exploring the links between natural products and bacterial assemblages in the sponge aplysina aerophoba. *Appl. Environ. Microbiol.*

Case studies

Mice models with Huntington's disease (HD data):

- Exploring differences in microbial composition between Huntington's disease and wild-type mice
- 368 microbial variables & 30 samples
- Effect of interest: 2 different genotypes
- (Biological) batch effect: samples from 10 different mouse cages

Cages\Genotypes	HD	WT
Cage A	2	0
Cage B	3	0
Cage C	2	0
Cage D	0	4
Cage E	0	4
Cage F	0	3
Cage G	3	0
Cage H	3	0
Cage I	2	0
Cage J	0	4

Kong, et al (2016). Microbiome profiling reveals gut dysbiosis in a transgenic mouse model of Huntington's disease. *Neurobiol Dis*.

Workflow for batch effect management

Input data (preprocessed)

I. Batch effect detection

II. Managing batch effects

Accounting for batch effects

Depends on the analytic models (e.g., p values for DE analysis)

Correcting for batch effects ★

Corrected data

III. Batch-effect correction assessment

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

I. Batch effect detection

Purpose: to detect batch effects and determine if the batch effect management is required.

A. Visual approaches: limited for very weak batch effects

- Principal component analysis (PCA)
- Boxplots and density plots
- Heatmap

B. Quantitative methods: very sensitive to batch effects

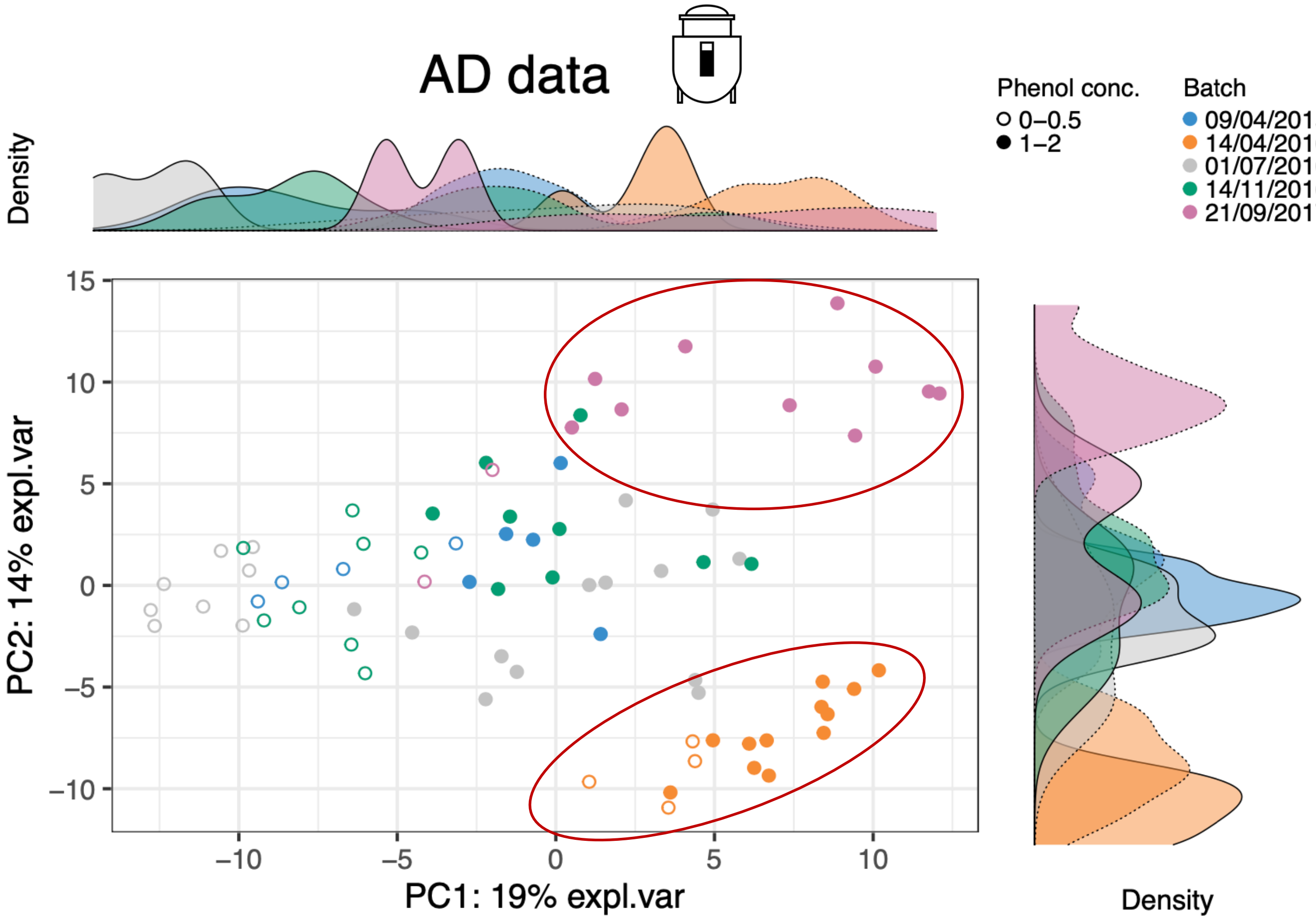
- Partial redundancy analysis (pRDA)

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

I. Batch effect detection

PCA plots with density per component

=> Multivariate: combination of all taxa



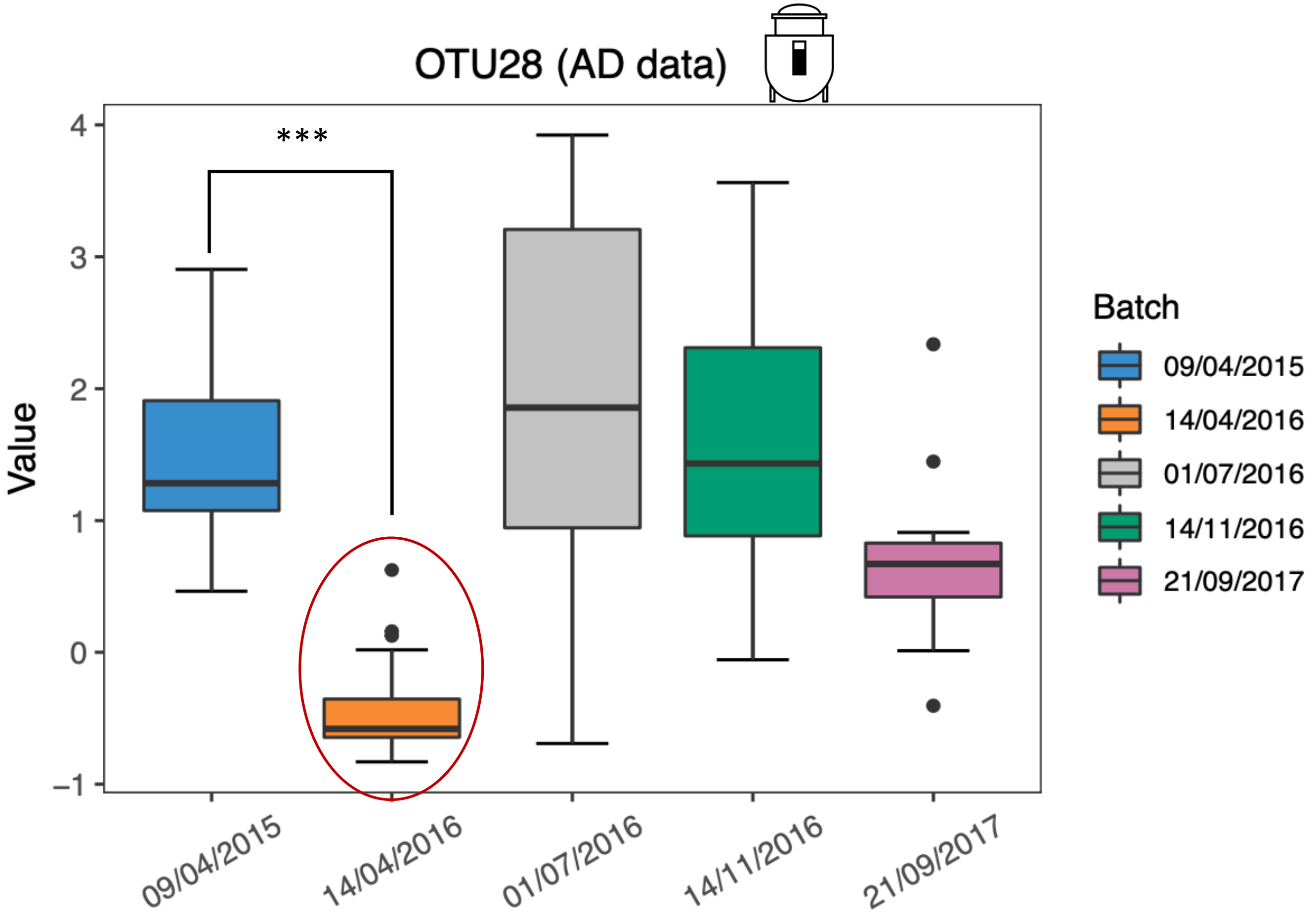
Batch variation is captured by the second PC.

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

I. Batch effect detection

Boxplots

=> Univariate: single taxon



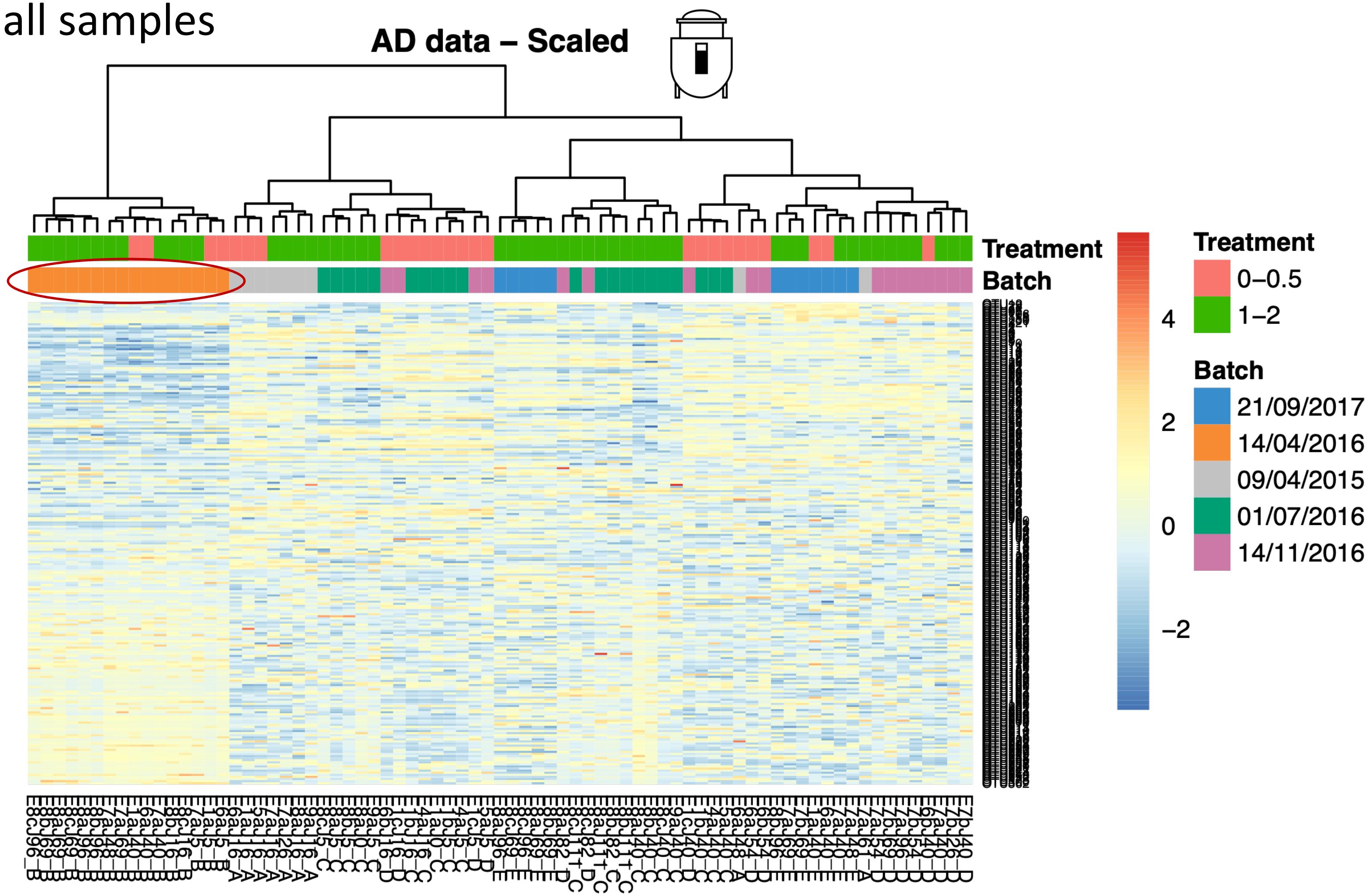
Test for the difference between batches (here P-value < 0.001, t-test).

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

I. Batch effect detection

Heatmap

=> All taxa and all samples



Samples from the batch dated "14/04/2016" clustered together.

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

I. Batch effect detection

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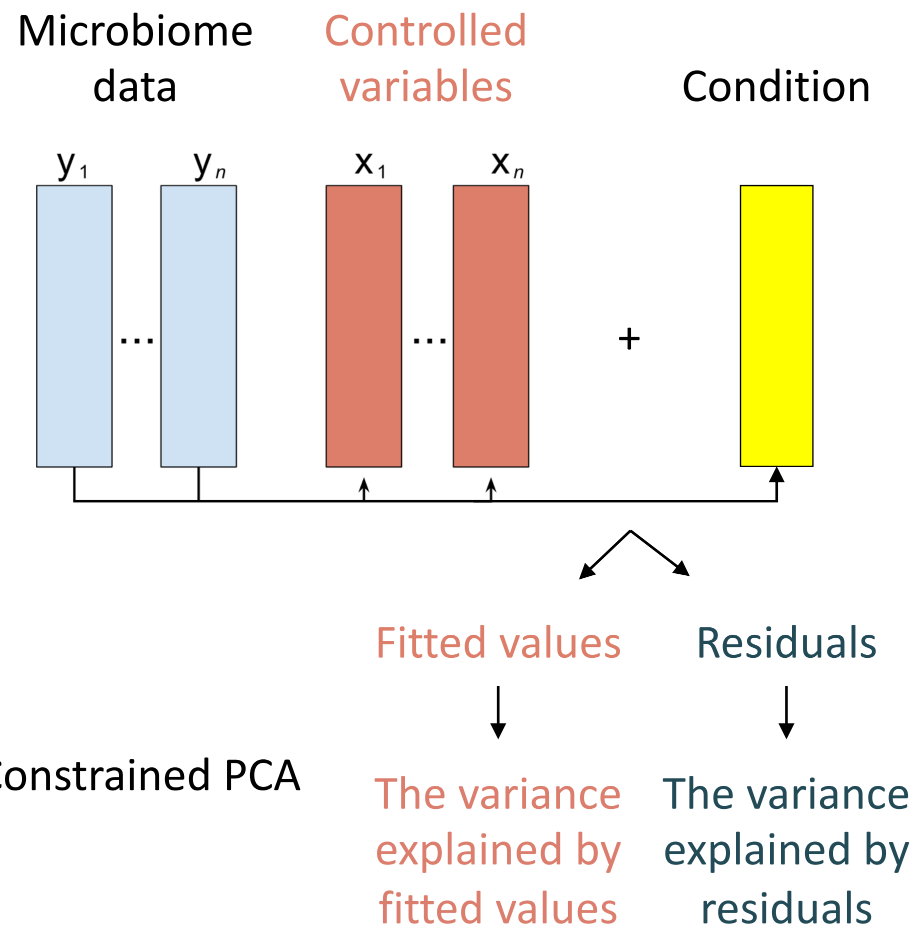
Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

I. Batch effect detection

Partial redundancy analysis (pRDA): multivariate

Multiple linear regression



Constrained PCA

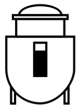
Unconstrained PCA

Output:
the proportion of variation in the microbial community explained by one factor after controlling for another factor, considering all microbial variables jointly.

Borcard, et al (1992). Partialling out the spatial component of ecological variation. *Ecology*.

I. Batch effect detection

pRDA: can indicate if batch x treatment design is balanced

Approx. balanced batch x treatment design (**AD data** )

Dates\Phenol conc.	0-0.5	1-2
09/04/2015	4	5
14/04/2016	4	12
01/07/2016	8	13
14/11/2016	8	9
21/09/2017	2	10

The intersection variance indicates how unbalanced the design is:

=> Batch and treatment effects are confounded.

##		Df	R.squared	Adj.R.squared	Testable
Treat only	= Treat | Batch	1	NA	0.08943682	TRUE
Intersection		0	NA	0.01296248	FALSE
Batch only	= Batch | Treat	4	NA	0.26604420	TRUE
## [d]	= Residuals	NA	NA	0.63155651	FALSE

I. Batch effect detection

pRDA:

Completely balanced batch x treatment design (sponge data)

	Tissue 1	Tissue 2
Batch 1	8	8
Batch 2	8	8

##		Df	R.squared	Adj.R.squared	Testable
Treat only	= Treat | Batch	1	NA	0.16572246	TRUE
Intersection		0	NA	-0.01063501	FALSE
Batch only	= Batch | Treat	1	NA	0.16396277	TRUE
## [d]	= Residuals	NA	NA	0.68094977	FALSE

No intersection variance:

=> Batch and treatment factors are independent.

I. Batch effect detection

pRDA:

Nested batch x treatment design (**HD data**)

Cages\Genotypes	HD	WT
Cage A	2	0
Cage B	3	0
Cage C	2	0
Cage D	0	4
Cage E	0	4
Cage F	0	3
Cage G	3	0
Cage H	3	0
Cage I	2	0
Cage J	0	4

##			Df	R.squared	Adj.R.squared	Testable
Treat only	=	Treat | Batch	0	NA	-2.220446e-16	FALSE
Intersection			0	NA	9.730583e-02	FALSE
Batch only	=	Batch | Treat	8	NA	1.608205e-01	TRUE
## [d]	=	Residuals	NA	NA	7.418737e-01	FALSE

No treatment variance & considerable intersection variance:

=> Batch and treatment factors are collinear.

Your turn!

Practice detecting batch effects in the AD data by following the steps in the "Batch effect detection" section. (30 mins)

II. Managing batch effects

- Accounting for batch effects:

  include batch factors as covariates in statistical models

- Correcting for batch effects:

  remove batch effects from the original data

- Univariate vs. multivariate:
  - Univariate: process each variable individually.

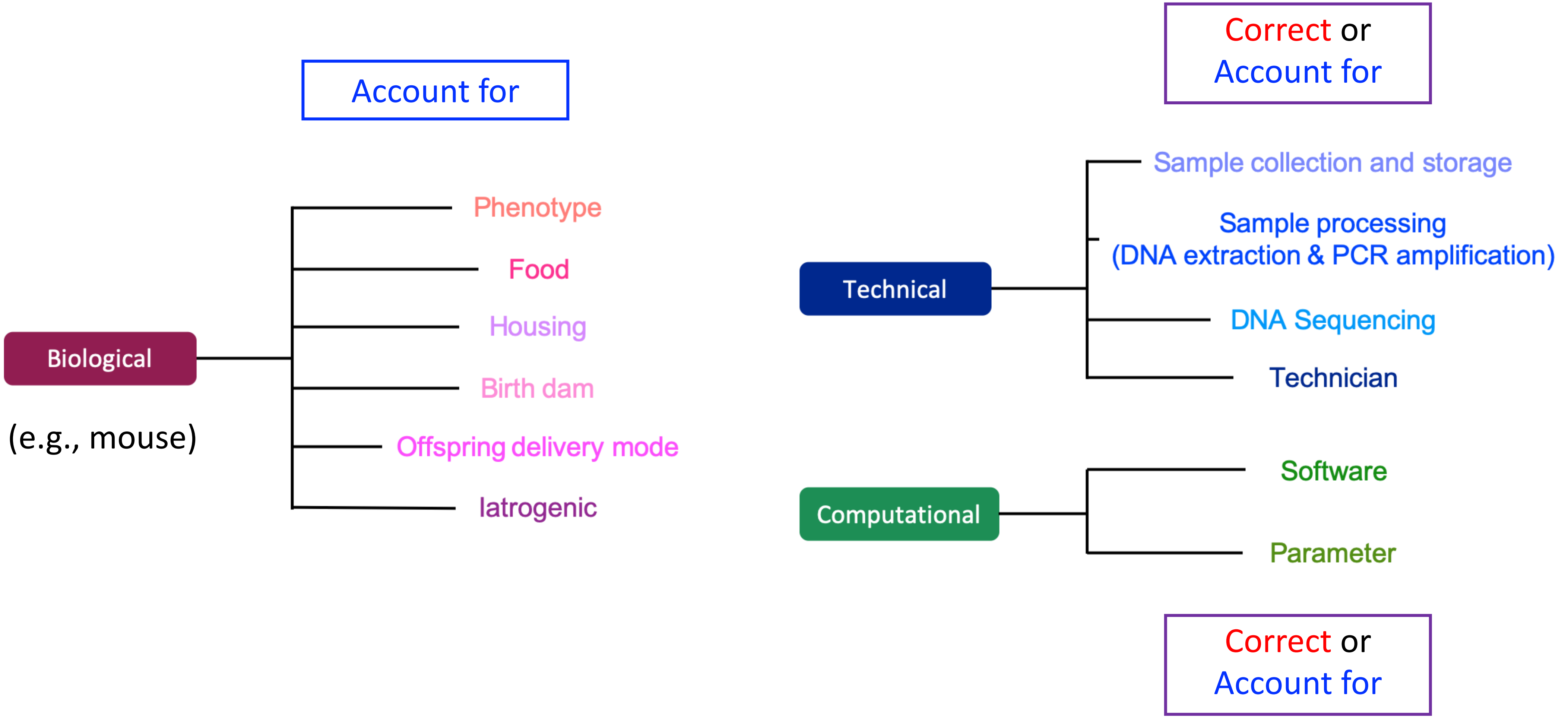
    E.g., Differential expression analysis. in *DESeq2*, *edgeR*
  - Multivariate: process all variables together.

    E.g., PCA, CCA, etc. in *phyloseq*

- Multivariate methods allow to consider variables as multivariate, rather than independent. E.g., better for microbiome data.

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

II. Managing batch effects



Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

II. Managing batch effects

Methods accounting for batch effects:

- Pros: can accommodate data characteristics and some degree of association between batch and treatment factors by including both in the model.
- Cons:
  - Limited to specific analyses according to the model (e.g., differential abundance analysis: taxa with p values)
  - Difficult to be assessed explicitly

e.g., Linear regression

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

II. Managing batch effects

Methods accounting for batch effects:

A. Designed for microbiome data (applied to count data):

- Cumulative-Sum Scaling normalisation + Zero-inflated Gaussian mixture model (CSS+ZIG):
  - Differential abundance analysis
  - Handle data characteristics, including uneven library sizes, compositional structure and undersampling zeroes

B. Adapted for microbiome data (after preprocessing):

- Linear regression: can handle nested batch x treatment design (**HD data**)
- Surrogate variable analysis (SVA): estimate unknown batch effects without extra information
- Remove unwanted variation in 4 steps (RUV4): estimate unknown batch effects but require negative control variables that capture the batch variation

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

II. Managing batch effects

Methods correcting for batch effects (main focus of this workshop)**:**

- Applied to preprocessed data, e.g., CLR transformed microbiome data

- Pros: corrected data can be input in many downstream analyses
  - Dimension reduction; Visualisation; Clustering; Network analysis; Data integration and prediction; But not recommended for variable selection

- Cons:
  - cannot account for specific data characteristics within models; these need to be addressed in advance, e.g., CLR transformation
  - cannot handle association between batch and treatment factors within models; require additional processing to consider it

e.g., ComBat

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

II. Managing batch effects

Methods correcting for batch effects:

- removeBatchEffect (rBE):
  - Linear regression (removeBatchEffect(), *limma*)
  - Univariate
- ★ ComBat:
  - Empirical Bayes method
  - Assumes all variables are affected by batch effects in a systematic manner
  - Feature-wise univariate modelling with cross-variable information sharing
- Percentile normalization (PN):
  - Each case sample's feature values are converted into percentiles of the control distribution within each batch
  - Requires sufficient control samples within each batch
  - Univariate

Wang & Lê Cao (2020). Managing batch effects in microbiome data. *Briefings in Bioinformatics*.

II. Managing batch effects

Methods correcting for batch effects:

- Remove Unwanted Variation-III (RUVIII):
  - Requires negative control variables and sample replicates that capture the batch variation
  - Multivariate

- PLSDA-batch:
  - Non-parametric: can handle non-Gaussian distributions
  - No assumption of systematic batch effects
  - Include two variants: sparse PLSDA-batch (avoid overfitting); weighted PLSDA-batch (for unbalanced batch x treatment design)
  - Multivariate

Wang & Lê Cao (2023). PLSDA-batch: a multivariate framework to correct for batch effects in microbiome data. *Briefings in Bioinformatics*.

Your turn!

Practice accounting for and correcting batch effects in the AD data by following the steps in the "Managing batch effects" section. (30 mins)

III. Assessing batch effect correction

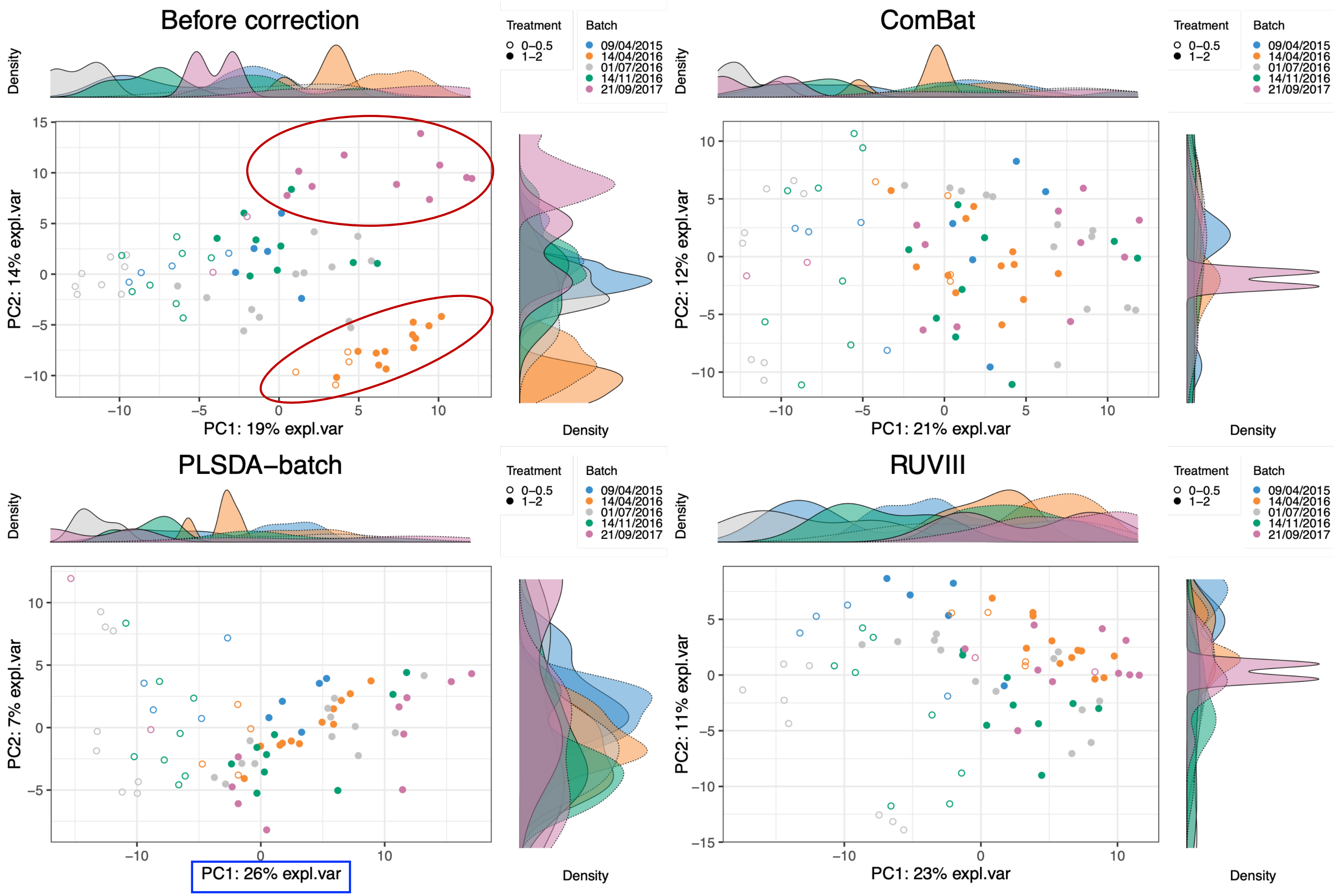
- **Methods that detect batch effects:**
  - Visualisation: PCA, boxplots, density plots, heatmap
  - pRDA: proportion of explained variance across all variables

- **Other methods:**
  - R^2 from one- way ANOVA: proportion of explained variance for each variable
  - Alignment scores: [0,1], poor to excellent mixing samples among the different batches

Wang & Lê Cao (2023). PLSDA-batch: a multivariate framework to correct for batch effects in microbiome data. *Briefings in Bioinformatics*.

III. Assessing batch effect correction

PCA:

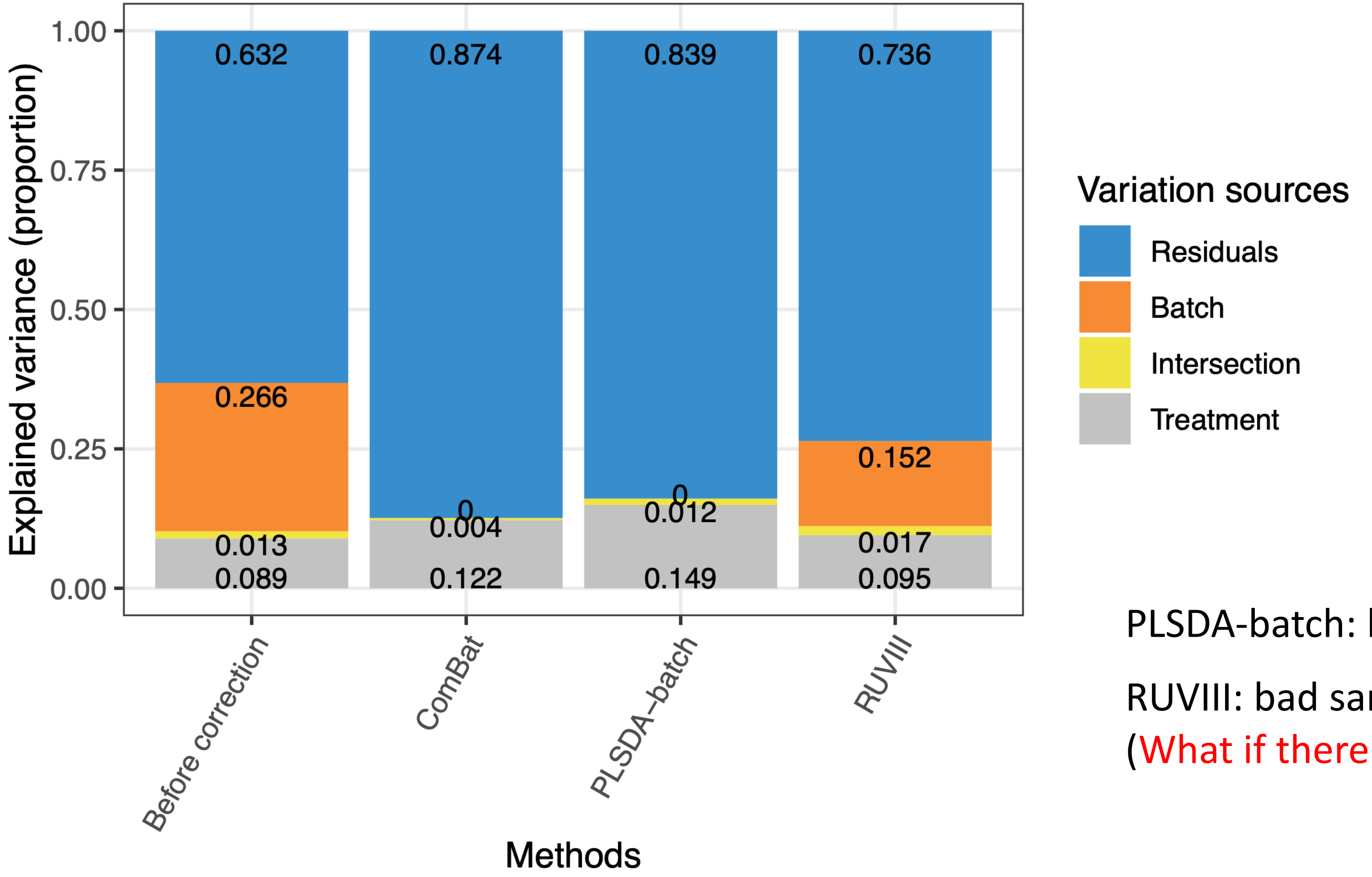


Batch variation is removed.

Wang & Lê Cao (2023). PLSDA-batch: a multivariate framework to correct for batch effects in microbiome data. *Briefings in Bioinformatics*.

III. Assessing batch effect correction

pRDA:



PLSDA-batch: higher treatment variance

RUVIII: bad sample replicates
(What if there were better replicates?)

Wang & Lê Cao (2023). PLSDA-batch: a multivariate framework to correct for batch effects in microbiome data. *Briefings in Bioinformatics*.

III. Assessing batch effect correction

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III. Assessing batch effect correction

Based on the sample dissimilarity matrix calculated from the PCA projection:

$$\text{Alignment scores} = 1 - \frac{\bar{x} - \frac{k}{n}}{k - \frac{k}{n}},$$

k: the number of nearest neighbours

n: the sample size

x: the number of each sample's nearest neighbours that belong to the same batch

$\bar{x}$: the average of all x

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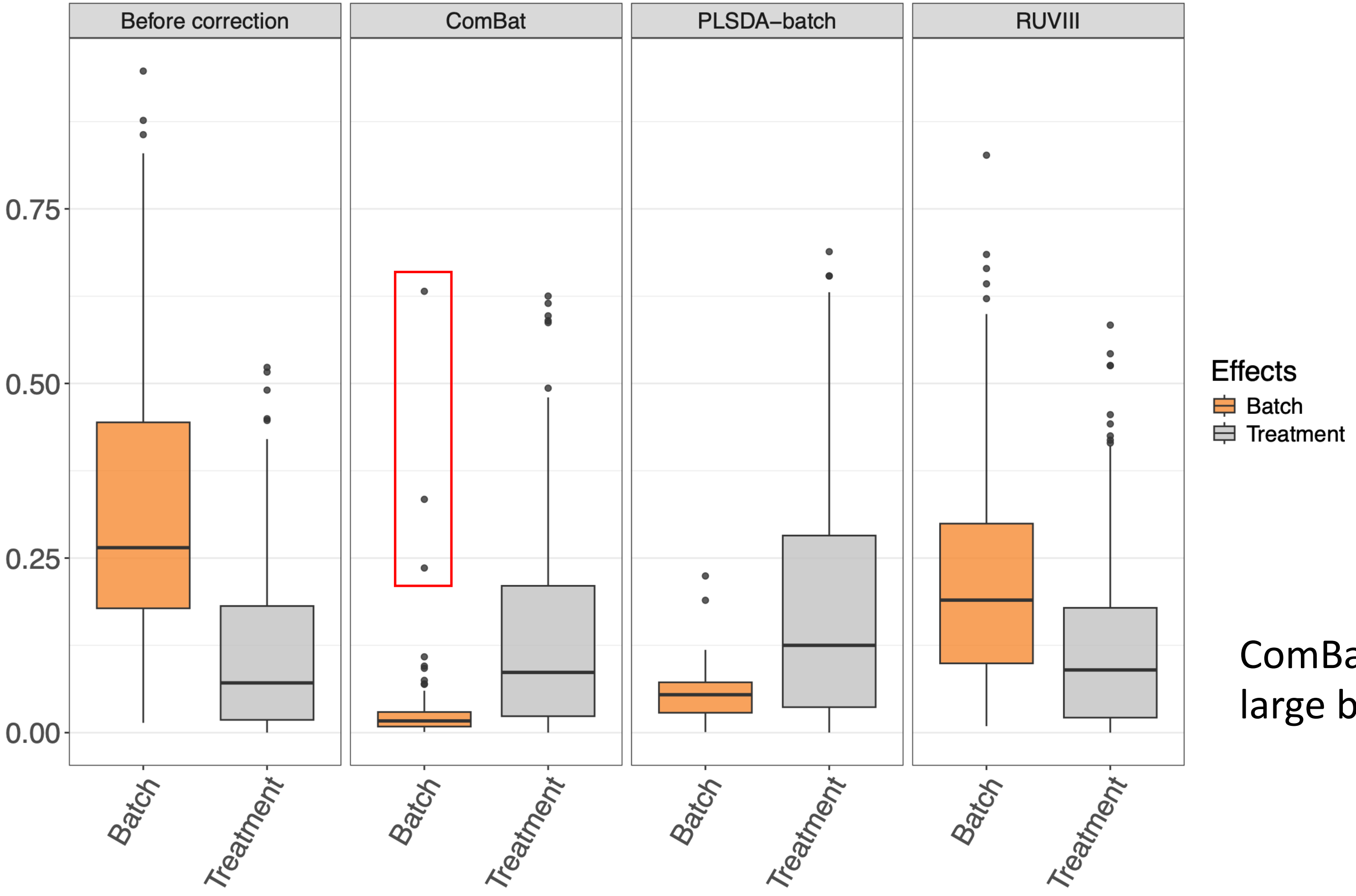
III. Assessing batch effect correction

R^2 from one- way ANOVA:

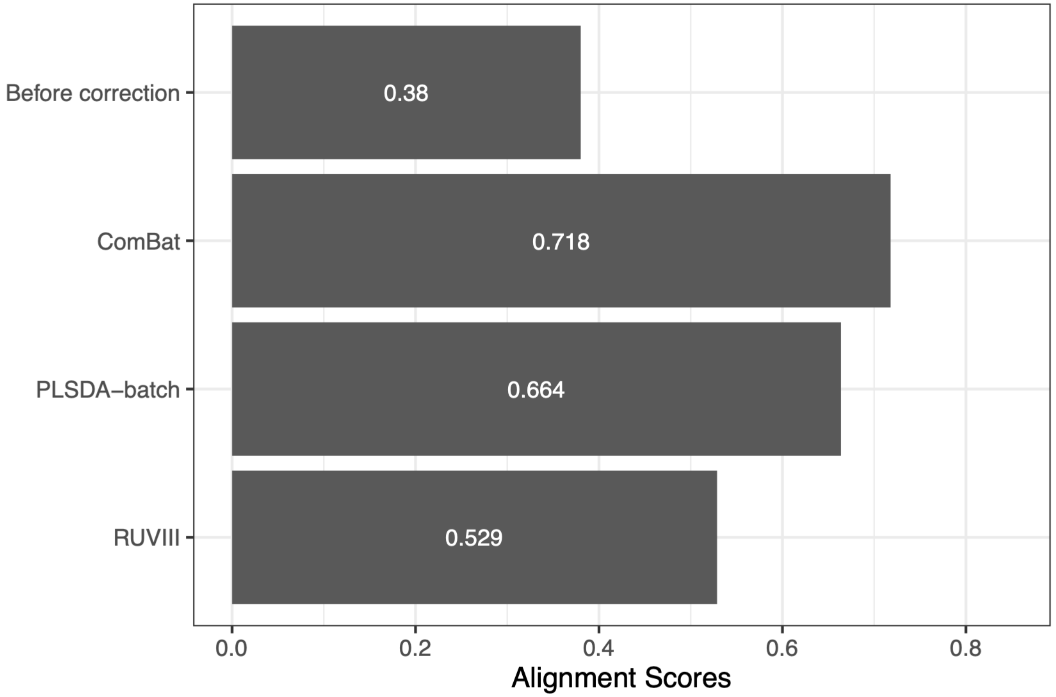


ComBat: a few variables with a large batch variance

Wang & Lê Cao (2023). PLSDA-batch: a multivariate framework to correct for batch effects in microbiome data. *Briefings in Bioinformatics*.

III. Assessing batch effect correction

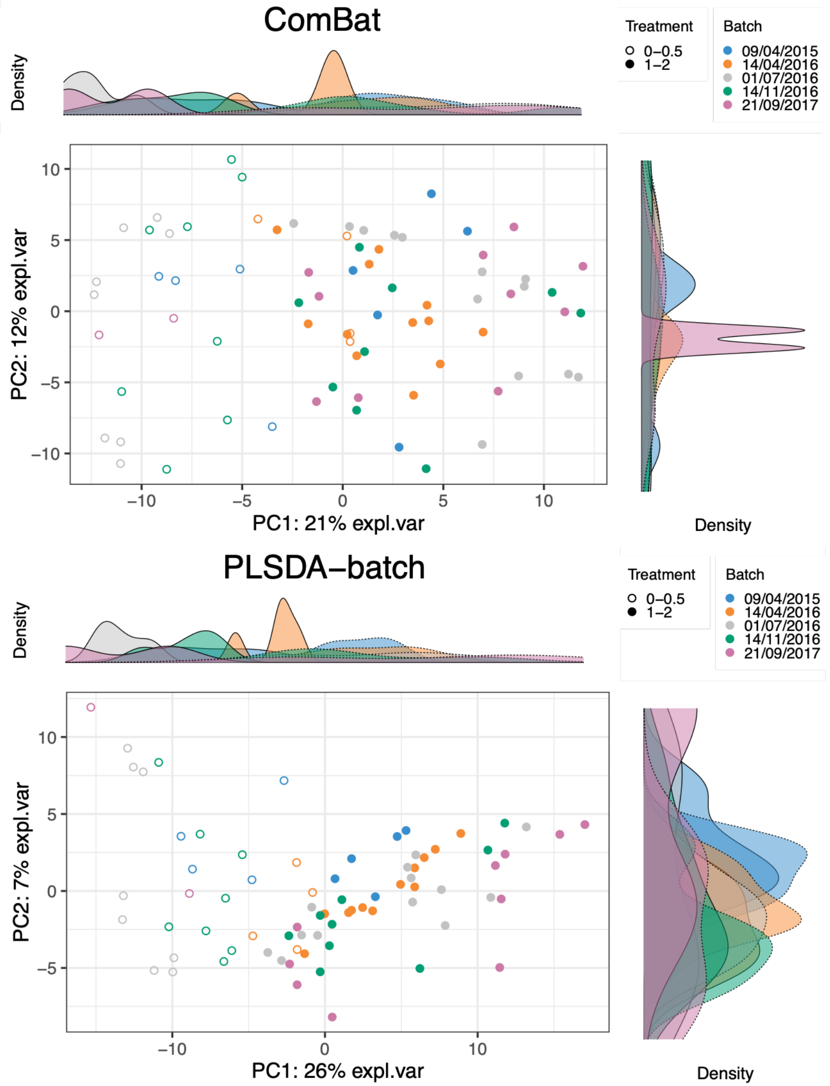
Alignment scores:



ComBat vs. PLSDA-batch :

Better mixing of batches?

→ Greater variance in PCA projection



Wang & Lê Cao (2023). PLSDA-batch: a multivariate framework to correct for batch effects in microbiome data. *Briefings in Bioinformatics*.

Your turn!

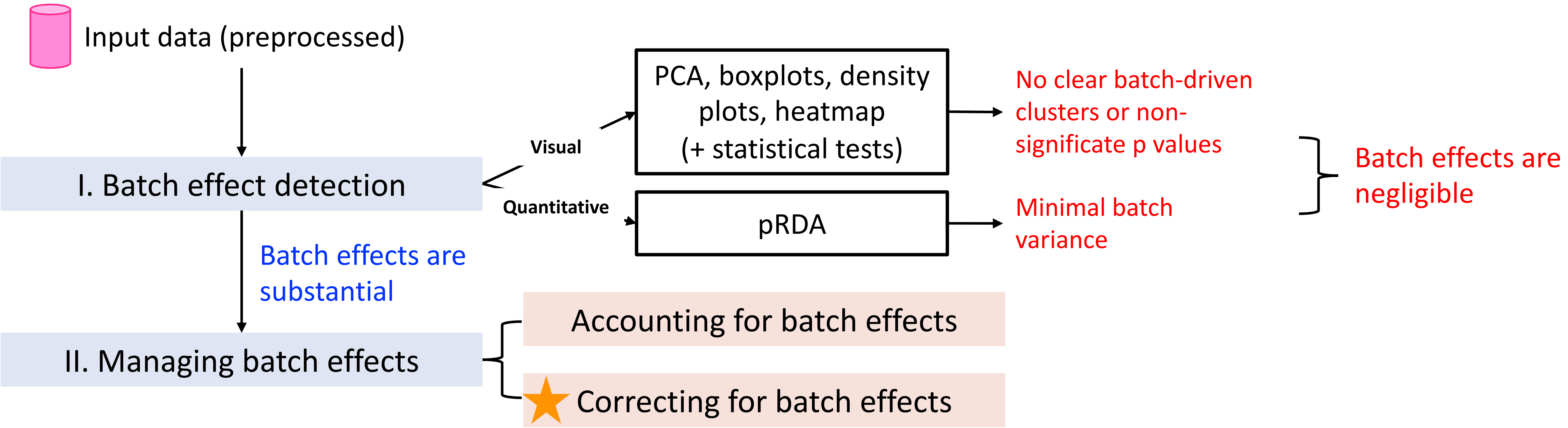
Practice using the AD data by following the steps in the "Assessing batch effect correction" section. (30 mins)

Conclusions

- Batch effects are ubiquitous and sometimes unavoidable. They represent unwanted variation in measured data that is not the primary focus of a study and may obscure the biological effects of interest.

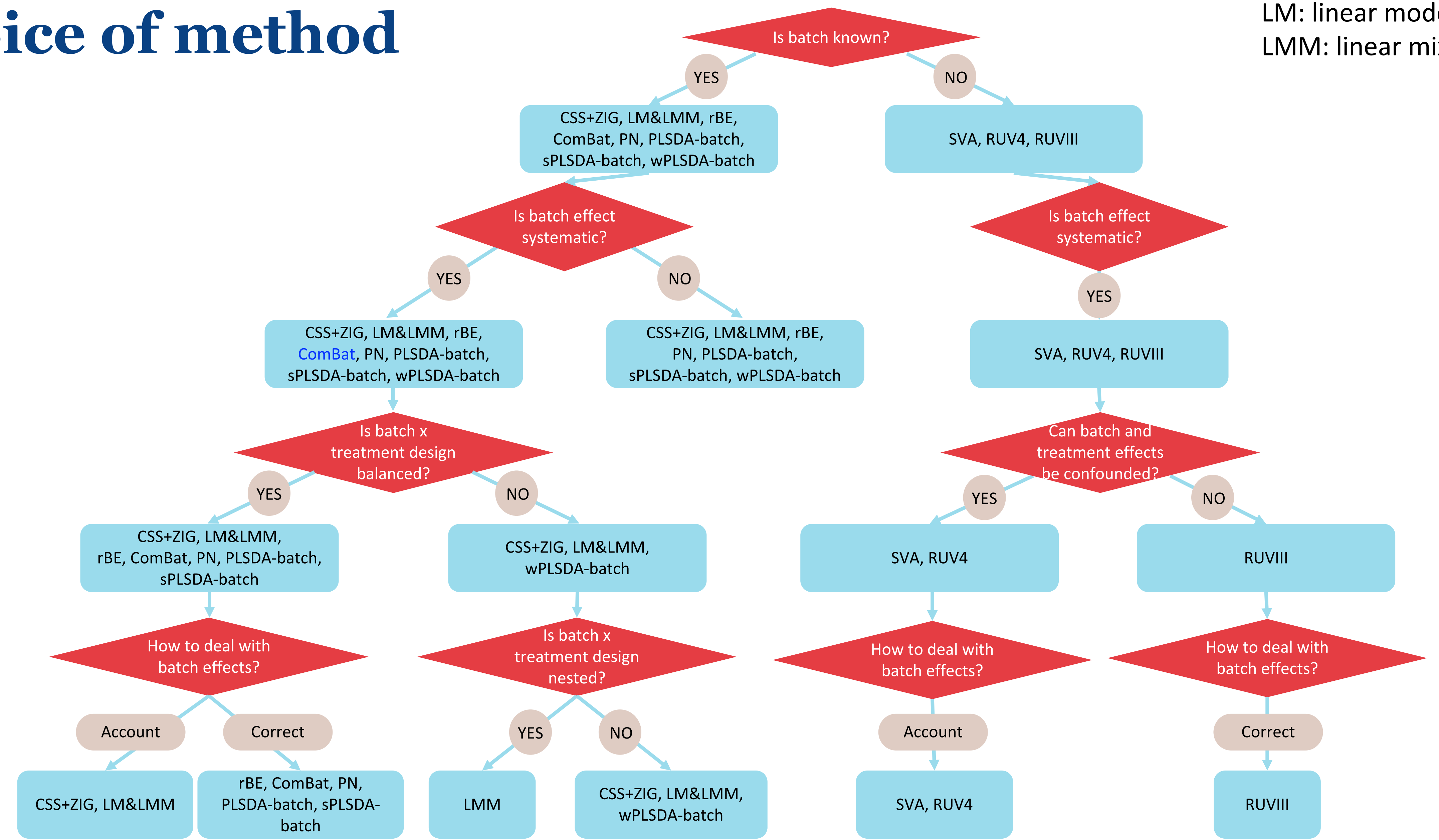
- Managing batch effects should consider:
  - Data characteristics: preprocessing beforehand or inclusion in the model
  - Batch sources: biological, technical and computational sources
  - Batch x treatment designs: balanced or confounded
  - Influence across variables: alignment with method assumptions

Conclusions

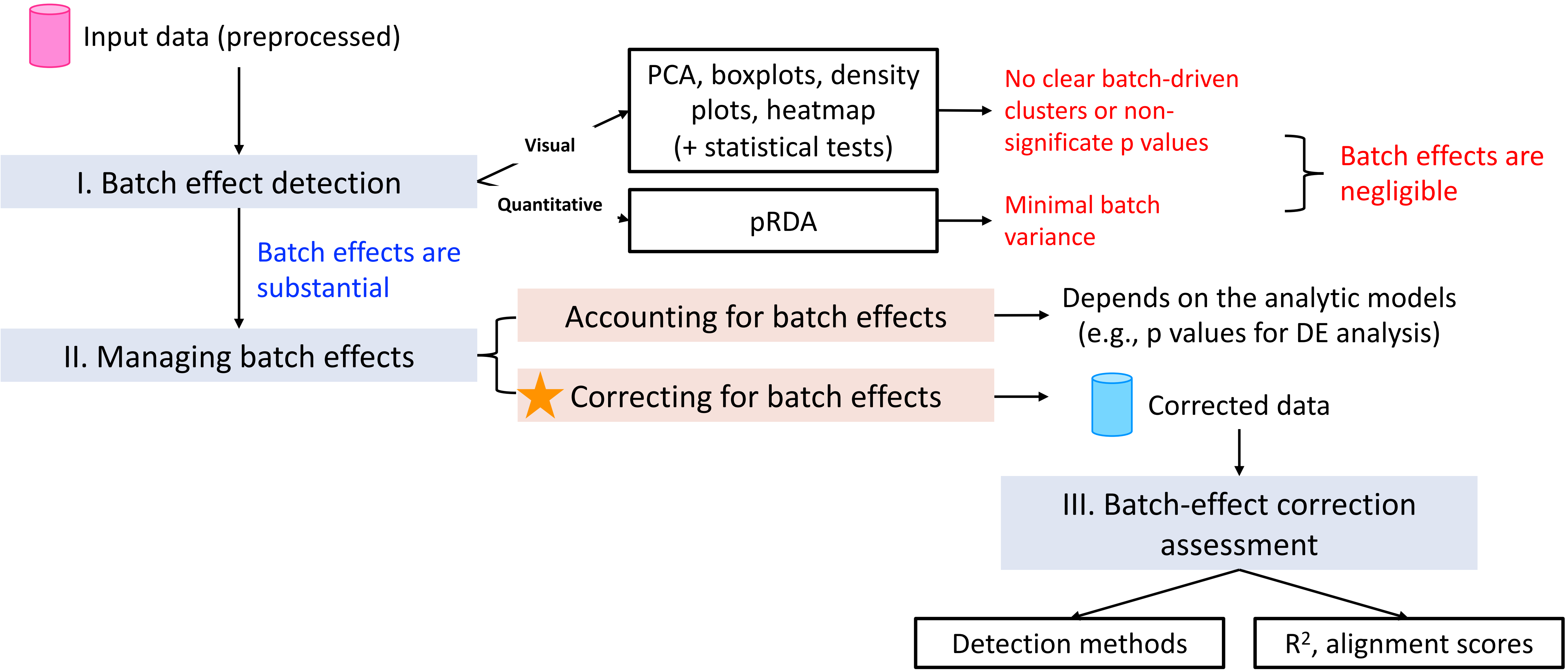


Choice of method

LM: linear model
LMM: linear mixed model



Conclusions



Next workshop

MIG Spatial Transcriptomics Workshop

Training or Workshop

Wednesday 5 August 2026 Add to my calendar

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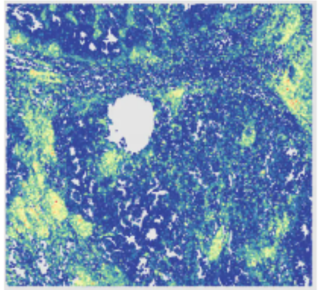

Kenneth Myer Building - Education Room (Ground Floor)

Date: Wednesday 5 August 2026

Time: 9:30am - 4:30pm

Host: Melbourne Integrative Genomics

Location: Kenneth Myer Building - Education Room (Ground Floor)

Cost: A registration fee applies. Please click '**Book Now**' to purchase your ticket.

Contact email

mig-ea@unimelb.edu.au

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About the Event

Spatial RNA-seq has enabled us to examine transcriptional activities of cells in complex tissue microenvironments. Maintaining the spatial context of cellular gene expression is vital for understanding how location influences cellular function and interaction within complex tissues. Despite these advantages, the field faces significant challenges in standardising analytical approaches. Current workflows for spatial RNA-seq data often involve fragmented pipelines with varying normalisation strategies, cell type identification methods and differential expression analyses.

Time to reflect and give feedback!

Please fill in the 3-question form before you leave!

It's really important for us to receive feedback so that we can continue delivering these workshops!

Appendix

Further information is available upon interest.

Data pre-processing for omics data

RNA-seq data:

Characteristics: Count data (discrete, non-negative), overdispersion, uneven library sizes, many low counts and zeros, high dimensionality

Normalisation:

- Trimmed Mean of M-values (TMM; *edgeR*)
- Median of Ratios size-factor normalisation (*DESeq2*)

Microbiome data:

Characteristics: Count data, high sparsity and a large proportion of zeros, overdispersion, uneven library sizes, compositional structure, high dimensionality and dependence among microbial variables

Normalisation and transformation:

- Centered Log-Ratio (CLR; *mixOmics*)
- Cumulative Sum Scaling (CSS; *metagenomeSeq*)

Data pre-processing for omics data

Metabolomic and proteomic data :

Characteristics: Typically continuous, non-negative intensity or abundance data, often right-skewed with a wide dynamic range, heteroscedasticity, missing values, high dimensionality and correlation among variables

Transformation:

- Filtering and imputation of missing values
- Log transformation to reduce skewness and stabilise variance
- Median or quantile normalisation to match distributions across samples
- Normalisation to internal / spike-in standards to control for sort of technical variation (systematic)